AMANI IBRAHIM CARBONELL

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Computer Engineering student with applied AI and computer vision research in Python and PyTorch. Developed two EfficientNet-B0 classifiers, curated multi-source image data, and evaluated performance using accuracy, precision, recall, and F1-score.

EDUCATION

University of South Florida | Tampa, FL

B.S. in Computer Engineering | Minor: Biomedical Physics | Expected May 2028

Polytechnic University of Puerto Rico | Computer Engineering coursework | **GPA: 3.6**

TECHNICAL SKILLS

Programming & Tools: Python, PyTorch, Torchvision, NumPy, C++, Java, SQL (MySQL), MATLAB

Machine Learning: Computer vision, image classification, CNNs, transfer learning, image augmentation, dataset preparation, model evaluation

RESEARCH EXPERIENCE

Undergraduate Researcher | Polytechnic University of Puerto Rico | 2025-2026

Honors and Outstanding Students (HOS) Research Program

- Achieved 94.07% best validation accuracy and 0.94 macro F1-score on a four-class emergency classifier by curating 1,949 labeled images from six public datasets and applying EfficientNet-B0 transfer learning.
- Developed two EfficientNet-B0 models in Python using PyTorch and Torchvision: a four-class emergency classifier and a binary damage-severity estimator.
- Reached 79.57% best validation accuracy on mild versus severe damage by applying class-weighted cross-entropy loss and staged fine-tuning to address class imbalance.
- Evaluated models using precision, recall, F1-score, and confusion matrices; authored a research paper documenting methodology, results, limitations, and future work.
- Designed a MySQL-compatible relational database schema linking image metadata to classification and severity predictions through foreign keys and model-version records.

PROJECTS

CineVault Movie Inventory | Java, Spring Boot, H2

Built a movie inventory web application with add, view, edit, and delete functions, movie-ID lookup, and duplicate validation.

ADDITIONAL EXPERIENCE

University Tutor (Work-Study) | Polytechnic University of Puerto Rico | 2025-2026

Tutored university students in calculus and physics, explaining concepts and guiding problem solving.

Front Desk Agent | Residence Inn by Marriott, Isla Verde | 2023-2024

Trained new staff and improved front desk workflows.

Front Desk Agent | Hyatt Regency Grand Reserve, Rio Grande | 2022-2023

Coordinated large-group arrivals and guest requests across departments.

Client Success Manager & Lead Sales Representative | BH Processing | 2018-2022

Signed more than 50 corporate clients and led a team supporting sales and customer service.

Languages: English, Spanish, Arabic (fluent)